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21L-5692

Q1 a) $(1234)_5$

$5^3 \times 1 + 5^2 \times 2 + 5^1 \times 3 + 5^0 \times 4$

$125 + 50 + 15 + 4$

(194

$$\begin{array}{r|l} 8 & 194 \\ 8 & 24 - 2 \\ & 3 - 0 \end{array}$$

Answer Octal

$\boxed{(302)_8}$

$$\begin{array}{r|l} 16 & 194 \\ & 12 - 2 \end{array}$$

Answer Hexa-decimal $\rightarrow$ $\boxed{(C2)_{16}}$

$$\begin{array}{r} 16 \\ \times 2 \\ \hline 32 \end{array}$$

Q2 a) $(234)_5$

$5^2 \times 2 + 5^1 \times 3 + 5^0 \times 4$

$50 + 15 + 4$

69

$$\begin{array}{r|l} 7 & 69 \\ 7 & 9 - 6 \\ & 1 - 2 \end{array}$$

$\boxed{(126)_7}$

Answer (a)

Range of base 5 $\rightarrow$ 32

$\rightarrow$ 0 − 31

Range of base 7 $\rightarrow$ 128

base 7 $\rightarrow$ 0 − 127

b) $(167)_5$

$5^2 \times 1 + 5^1 \times 6 + 5^0 \times 7$

$25 + 30 + 7$

62

$$\begin{array}{r|l} 7 & 62 \\ 7 & 8 - 6 \\ & 1 - 1 \end{array}$$

$\boxed{(116)_7}$ $\leftarrow$ Ans

Q3) a 8 - 14

$(1000)_2 - (1110)_2$

Taking 2's compliment of negative number

	1110
1's comp	0001
Add 1	+ 1
2's comp	0010

Adding numbers	1000
	+ 0010
	1010

Take 2's compliment of Answer

	1010
1's comp	0101
Add 1	+ 1
	0110

Final Answer - 0110 or - 6

b -6 - 17

-00110 - 10001

00110	Taking 2's Compliment	10001
11001 ←		→ 01110
+1		+1
11010		01111

$$\begin{array}{r} 11010 \\ +\ 01111 \\ \hline \boxed{1}\,01001 \end{array}$$ ← Addition

Two's Compliment of result

01001

10110

+1

-10111

Final Answer -10111 or -23

c) 15 - 6

1111 - 0110

Two's compliment of negative number

0110
1001
+1
1010

Adding numbers

1010
1111
$\boxed{1}$ 1001

Final Answer 1001 or 9

Q4
```
   111
 × 101
 ------
   111
  000x
 111xx
 ------
1 00011
```
Answer → 1 00011

Q5 a)
```
8421 8421 8421 8421
1100 1010 0101 0111
 12   10    5    7
  C    A    5    7
```
$\boxed{(CA57)_{16}}$ ← **Ans**

~~b) 001100 1010~~

b)
```
8421 8421 8421 8421
0011 0010 1010 1001
  3    2   10    9
```
$\boxed{(32A9)_{16}}$ ← **Ans**

Q6 a) 825
```
  8      2      5
8421   8421   8421
1000   0010   0101
```
$\boxed{(1000\ 0010\ 0101)}$ ← **Ans**

b) 709
```
  7      0      9
8421   8421   8421
0111   0000   1001
```
$\boxed{(0111\ 0000\ 1001)}$ ← **Ans**

PAPERSMITH

Q7) a)

1	0	A
8 4 2 1	8 4 2 1	8 4 2 1
0 0 0 1	0 0 0 0	1 0 1 0

$\boxed{(0001\,0000\,1010)_2}$ ← Ans

b)

2	B	C
8 4 2 1	8 4 2 1	8 4 2 1
0 0 1 0	1 0 1 1	1 1 0 0

$\boxed{(0010\ 1011\ 1100)_2}$ ← Ans

Q8) a) $16^1 \times 14 + 16^0 \times 5$

$224 + 5$

$\boxed{(229)_{10}}$ ← Ans

16×14: 64, 16x, 224

Q9) a) (B)

1 F
× C
$\boxed{174}_{16}$

Ans

16) 23 (1: 16, 7

15 × 12: 30, 15x, 180

16) 180: 16, 20, 16, 4

16 × 11: 16, 16x, 176

b) ~~(3B)~~

① ③
② ⑧

2 3 B
× 5 A
~~[illegible]~~
1 6 4 E
B 2 7 x

$\boxed{(C8BE)_{16}}$ ← Ans

16 × 8 = 128

16 × 7 = 112

16) 36 (2: 32, 4

16) 55 (3: 48, 7

PAPERSMITH

Q10 a) $(162)_8 + (537)_8$

$$\begin{array}{r} 162 \\ +\ 537 \\ \hline \boxed{(721)_8} \end{array}$$

← Ans

$$\begin{array}{r} 8\overline{)9} \\ -8 \\ \hline 1 \end{array} \qquad \begin{array}{r} 8\overline{)10} \\ -8 \\ \hline 2 \end{array}$$

b) $(136)_8 + (636)_8$

$$\begin{array}{r} 136 \\ +\ 636 \\ \hline \boxed{774} \end{array}$$

← Ans

Q11 a)

$$\begin{array}{r} 1011011 \\ \times 10 \\ \hline 0000000 \\ 1011011\times \\ \hline \boxed{(10110110)_2} \end{array}$$

← Answer

b)

$$\begin{array}{r} 431022 \\ \times 10 \\ \hline 000000 \\ 431022\times \\ \hline \boxed{(4310220)_5} \end{array}$$

← Answer

Q12)

A	B	C	Even X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

b)

A	B	C	Odd Bit Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Q13)

```
          1 0 1 1 . 0 1
        x   1 1 0 . 1
        1 0 1 1 0 1
      0 0 0 0 0 0 X
    1 0 1 1 0 1 X X
  1 0 1 1 0 1 X X X
1 0 0 0 0 0 1 0 0 1
```

$\boxed{1}$ 000001.001 ← Answer

```
            1 0 1 1 0 1 . 0 1 1
          x 1 1 0 . 1 0 1
            1 0 1 1 0 1 0 1 1
          0 0 0 0 0 0 0 0 0 X
        1 0 1 1 0 1 0 1 1 X X
      0 0 0 0 0 0 0 0 0 X X X
    1 0 1 1 0 1 0 1 1 X X X X
  1 0 1 1 0 1 0 1 1 X X X X X
1 0 0 0 0 0 0 1 1 . 1 0 0 1 1 1
```

$\boxed{1}$ 00000011.100111 ← Answer